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Research Article

Text-augmented long-term relation dependency learning for knowledge graph representation

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ABSTRACT

Knowledge graph (KG) representation learning aims to map entities and relations into a low-dimensional representation space, showing significant potential in many tasks. Existing approaches follow two categories: (1) Graph-based approaches encode KG elements into vectors using structural score functions. (2) Text-based approaches embed text descriptions of entities and relations via pre-trained language models (PLMs), further fine-tuned with triples. We argue that graph-based approaches struggle with sparse data, while text-based approaches face challenges with complex relations. To address these limitations, we propose a unified Text-Augmented Attention-based Recurrent Network, bridging the gap between graph and natural language. Specifically, we employ a graph attention network based on local influence weights to model local structural information and utilize a PLM based prompt learning to learn textual information, enhanced by a mask-reconstruction strategy based on global influence weights and textual contrastive learning for improved robustness and generalizability. Besides, to effectively model multi-hop relations, we propose a novel semantic-depth guided path extraction algorithm and integrate cross-attention layers into recurrent neural networks to facilitate learning the long-term relation dependency and offer an adaptive attention mechanism for varied-length information. Extensive experiments demonstrate that our model exhibits superiority over existing models across KG completion and question-answering tasks.

1. Introduction

Knowledge Graphs (KGs) are graph-structured semantic networks consisting of entities and relations, supporting a wide range of applications, such as recommendation [1], intelligent search [2], online analytical processing [3] and question answering (QA) [4]. KG representation learning, also denominated as knowledge graph embedding (KGE), which aims to learn the low-dimensional representations of entities and relations, is the foundation of current neural models to implement the aforementioned KG-driven applications.

Existing KG representation learning methods can be classified into the following two categories: (1) Graph-based methods, such as RotatE [5] and TransE [6], map entities and relations into vector spaces, leveraging the structural information to learn their representations. (2) Text-based methods like SimKGC [7] utilize additional textual information to fine-tune the BERT [8] for KGE. However, both types of methods suffer from the following two issues.

(a) Inadequate exploitation of both structural and textual information. Graph-based methods demonstrate efficient structural modeling capabilities but overlook the textual semantic connections between elements. Additionally, these methods are impacted by the incompleteness inherent in KGs. For entities and relations with fewer relevant triples, these methods struggle to learn effective representations [9]. In contrast, text-based methods that leverage textual corpus exhibit superior generalizability and robustness, but they tend to overlook the complex structural information in KGs, making it challenging to model KGs with numerous complex relations [7]. Besides, the inherent differences between graph structures and natural language poses challenges in integrating KGE methods into downstream tasks, such as Knowledge Graph Completion (KGC) and Knowledge Graph Question Answering (KGQA) . KGC refers to the task of inferring missing facts in a KG by predicting missing links between entities, while KGQA involves answering natural language questions by retrieving relevant facts from a KG. As illustrated in Fig. 1, questions, typically presented in natural language, require the retrieval of answers from a graph-structured KG. Previous methods have not adequately addressed the simultaneous handling of natural language and complex relational structures, leading to

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difficulties in understanding the semantic relevance between relations such as *co-founder* and *founded*, or *belong*, *serve*, and *technical support*. This oversight creates barriers to solving problems that require both graph structures and natural language.

(b) Low expressiveness for multi-hop semantic information. The major KGE methods focus on the triple-level and show poor performance on multi-hop reasoning tasks [10] which is a widely prevalent and challenging problem. Besides, different datasets and tasks have flexible requirements for the length of reasoning [11]. As shown in Fig. 1, answering the question necessitates the utilization of both short-term relation *co-founder* and long-term relation *belong* and *serve*. Some works such as RSN [10] and A* [12] are dedicated to modeling multi-hop paths, but they overlook the knowledge within individual triples [11]. Efficiently modeling multi-hop paths while adaptively focusing on information of varying lengths remains unexplored.

To jointly utilize the structural information of the KG and the textual information of its corresponding text corpus, as well as to adaptively model multi-hop semantic information, we propose a novel **TAARN**: Text-Augmented Attention-based Recurrent Network. To address limitation (a), we introduce a method for measuring local influence between entities and efficiently capturing local structural information in the KG with Graph Attention Networks (GAT). Simultaneously, our model employs prompt learning techniques to mitigate potential embedding biases in PLMs, thereby improving the representation of textual information. Additionally, we consider both the local influence function and the degree of each entity and implement a structural mask-reconstruction strategy based on global influence weights. By masking and then reconstructing important triples, our model effectively captures more significant structural information, leading to more robust representations. Furthermore, by leveraging textual contrastive learning techniques, our model significantly enhances the generalization of PLMs across various tasks.

To address limitation (b), we propose a novel semantic-depth guided path extraction algorithm, harnessing the capabilities of a biased random walk strategy based on semantics and depth to distill multi-hop paths from the KG. By leveraging the prior semantic knowledge within PLM and the structural information from KGs, we focus on challenging and deeper samples during the path extraction, and model explicitly these paths via Recurrent Neural Networks (RNNs), which improve the model's training efficiency and substantially boosts the effectiveness of information transference during complex multi-hop inferences. Besides, we integrate a cross-attention layer to dynamically focus on distinct node-specific data along the multi-hop path.

Contributions are summarized as follows:

1. We bridge the gap between graph and natural language and propose a novel representation model TAARN. Specifically, we utilize a GAT based local influence weights to capture local structural information and employ a PLM based on prompt learning to learn textual information. Additionally, we enhance the robustness and generalizability by utilizing a structural mask-reconstruction strategy and textual contrastive learning method.
2. To effectively model multi-hop semantic information, we propose a novel semantic-depth guided path extraction algorithm to extract higher quality relation paths and then utilize an RNN with cross-attention for explicit modeling, which captures the long-term relation dependency and exhibits length-adaptive attention for different tasks.
3. Extensive experiments demonstrate that TAARN manifests a superiority over existing models across KGC and KGQA tasks, and verify our model's efficient modeling and integration capabilities for both structural and textual information. Additionally, achieving SOTA performance across various datasets and tasks further confirms the generalizability of our model.

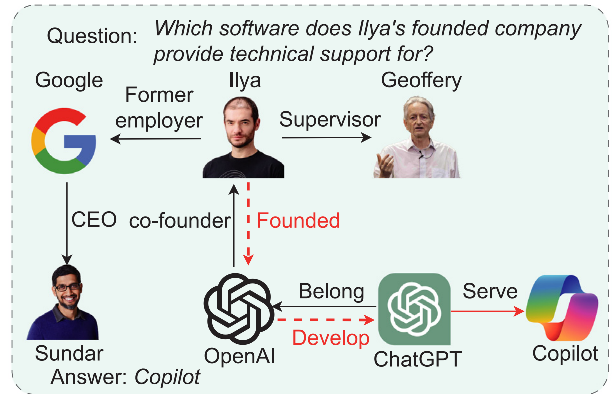

Fig. 1. A knowledge graph related to *Ilya Sutskever*. The red line means the inference path for the question. The dashed line represents relations that do not exist but can be inferred based on existing semantic information.

2. Related work

2.1. Knowledge graph representation learning

Graph-based and text-based methods are two main methods for KGE. Graph-based approaches [5,6,13] initially encode graph elements into dense vectors and then design a score function to measure the plausibility of triples in vector spaces, enabling the updating of vectors for entities and relations to learn representations of the KG. TransE [6] proposes an efficient KG embedding method by viewing a relation as the translation of two entities in a specific space. RotatE [5] and TransH [13] improve TransE's ability to learn various relation patterns and complex multi-mapping relations. In addition, some approaches [14,15] represent the relation-based interaction of the triples via linear operations or bilinear operations. However, these models ignore the multi-hop paths in a KG. RSN [10] seeks to overcome this shortcoming by utilizing an improved RNN for modeling the given KG, but ignoring the information inside a single triple. Text-based approaches [7,16,17] mainly exploit PLMs to obtain textual information corresponding to entities and relations. KG-BERT [16] employs the BERT [8] model to encode the corresponding text of fact triples through the cross-encoder architecture. Recently SimKGC [7] employs Bi-encoder to bridge the gap between PLM and KGC. These methods take more advantage of the knowledge in PLMs but do not pay sufficient attention to the structures of KGs, and thus achieve poor performance.

2.2. Knowledge Graph Completion

KGE-based approaches [5,6,18] strive to model a robust low-dimensional representation of KGs, utilizing vectors of entities and relations to predict missing information. Nevertheless, a majority of such approaches primarily draw upon triple-level knowledge, manifesting a deficiency in efficiently leveraging multi-hop information. Path-based approaches search for paths between the entities in a KG to infer paths and completion. While statistic-based path ranking algorithms [19] were the main methods utilized in the past, recent work [20] exploited reinforcement learning to explore multi-hop paths. Rule-based approaches attempt to utilize logic rules mining tools such as DRUM [21] and RLVLR [22] to obtain rules for KGC. However, both types of approaches are time-consuming and difficult to apply to other tasks solely from the perspective of KGC.

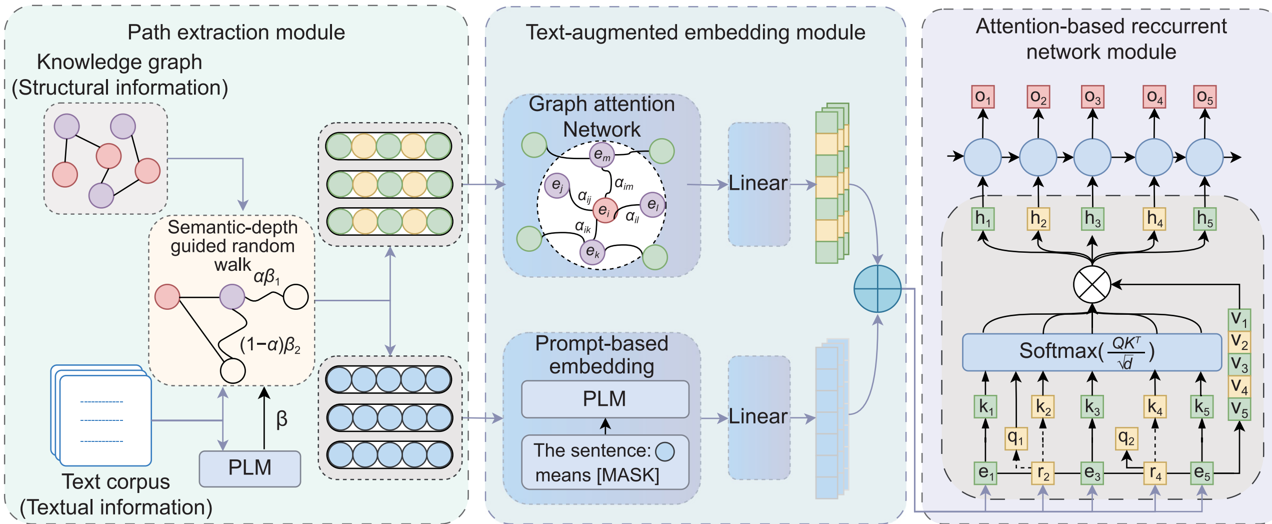

Fig. 2. Overview of TAARN. **Path extraction module**, **text-augmented embedding module**, and **attention-based recurrent network module** are three core components of TAARN. **Path extraction module** utilizes the biased random walk method to extract paths from a KG. Due to the one-to-one correspondence between entities, relations, and their text descriptions, we employ a PLM to encode the text descriptions, thereby augmenting the structural information. **Text-augmented embedding module** embeds the structural and textual information separately and fuses them. Finally, we employ the **attention-based recurrent network** to learn knowledge of varying lengths within the KG.

2.3. Question answering over knowledge graph (KGQA)

The purpose of KGQA is to find answers from KGs using natural language questions. Information extraction methods represent questions through templates or neural networks to identify candidate answers. Some methods [23,24] build subgraphs for entities or questions, using embeddings or additional text for answer extraction. However, challenges arise in defining subgraph size and retrieving longer entities. Other approaches [25,26] learn relation-question similarities or convert QA into Knowledge Graph Completion (KGC) tasks using pre-trained models. Yet, their reliance on triple-based KGC models limits their effectiveness in multi-hop queries. Semantic parsing-based methods offer an alternative, parsing tree-structured KG queries from input questions. Some methods [27,28] provide interpretability but face high time complexity and are constrained by the KG's completeness. Recent works [29,30] enhance multi-hop reasoning and logical querying on KGs through learned logical query embeddings. However, their integration with natural language queries remains a developing area. Both paradigms excel in simple queries but struggle with complex relations and multi-hop paths in KGQA.

3. Our approach

We provide a detailed introduction to our proposed model TAARN in this section. **Path extraction module**, **text-augmented embedding module**, and **attention-based recurrent network module (ARN)** are three core components of TAARN. Additionally, we also detail the implementation of the model learning. Specifically, the structural mask-reconstruction strategy, the textual contrastive learning method, and the core loss function. Fig. 2 shows the overall architecture.

3.1. Preliminaries

KG can be indicated as $\mathcal{G} = (\mathcal{E}, \mathcal{R}, \mathcal{T})$, where $\mathcal{E}$ denotes the set of entities, $\mathcal{R}$ denotes the set of relations, and $\mathcal{T}$ denotes the set of all existing triples. Moreover, the textual information used in KGC tasks refers to the text corpus of the KG, including text descriptions for each entity and relation. In KGQA tasks, the textual information also encompasses the natural language present in the questions, which also can similarly be considered as text descriptions of relations.

3.2. Path extraction module

The path extraction module extracts paths from the KG to enable the ARN to learn long-term dependencies. Limiting path extraction to avoid excessive computational complexity and inefficiency, focusing on constrained sampling techniques such as random walk, proves more practical than exhaustive path extraction in large-scale KGs. DeepWalk [31] utilizes the conventional unbiased random walk algorithm to draw a fixed length from the graph.

$$P(n_{i+1} = v \mid n_i = u) = \begin{cases} \frac{\pi_{uv}}{Z} & v \in N_{+}(u) \\ 0 & \text{otherwise} \end{cases} \tag{1}$$

where n_i represents the current node, and n_{i+1} represents the candidate node for the next sampling step in the path extraction process. Moreover, $N_{+}(u)$ denotes the set of all entities that are adjacent to entity u, π_{uv} denotes the transition probability of entity u to v and Z is a normalization constant.

Empirically, each triple in the KG holds varying impact weights for the effective representation. Specifically, triples that are simple to infer require only minimal training to achieve a good representation, while those that are difficult to infer necessitate more extensive learning, making them relatively more important for representation learning. Besides, longer paths provide richer long-sequence semantic knowledge for KG representation learning, which is crucial for handling complex multi-hop reasoning tasks.

In this paper, we propose a novel semantic-depth guided path extraction algorithm, harnessing the inherent semantic priors within PLMs and the structural information from KGs to implement both semantic and depth biases. Guided by these biases, the random walk algorithm can identify and sample more significant information, facilitating the exploration of deeper paths, thereby optimizing training efficiency and enhancing the model's ability to represent complex multi-hop information structures.

Specifically, we employ a PLM for encoding and measuring the importance of triples based on the textual similarity of their text descriptions to calculate semantic bias. We first concatenate the text descriptions of the head entity h and the relation r using the [SEP] token, which is a special token used in PLMs to separate different segments of input text, and employ a PLM to embed them, obtaining the embedding vector e_{hr}. Subsequently, we use the PLM to embed the text description of the tail entity t, resulting in embedding vector e_t. The semantic similarity between the two embedding vectors e_{hr} and e_t is calculated using the following formula:

$$sim(e_{hr}, e_t) = \cos(e_{hr}, e_t) = \frac{e_{hr} \cdot e_t}{\|e_{hr}\| \; \|e_t\|} \tag{2}$$

Based on our hypothesis, a high degree of initial textual similarity in a triple suggests a straightforward prediction, warranting a reduced emphasis on such triples in the path extraction. On the other hand, a lower initial textual similarity indicates a more complex predictive scenario, necessitating a greater focus on these triples during path extraction. To align with this reasoning, we derive our semantic bias term using a specific computational formula:

$$\pi^s_{n_i n_{i+1}} = \beta_{n_i n_{i+1}} = 1 - sim(e_{n_i r}, e_{n_{i+1}}) \tag{3}$$

where n_i represents the current node, and n_{i+1} represents the candidate node for the next sampling step in the path extraction process. Moreover, $\pi^s_{n_i n_{i+1}}$ represents the semantic bias between nodes n_i and n_{i+1}, $sim(e_{n_i r}, e_{n_{i+1}})$ denotes the textual similarity between the embedding $e_{n_i r}$ and the embedding $e_{n_{i+1}}$. This formula calculates the semantic bias as the complement of the textual similarity, ensuring that lower similarity pairs receive higher bias values. By incorporating this semantic bias, our model can adaptively focus on more challenging predictive scenarios, potentially improving its overall performance and generalization capability.

In addition, we introduce a depth bias to control the path depth [10]. Assuming that n_i denotes the current node, n_{i-1} denotes the previous node in the path, and n_{i+1} denotes the candidate node for the next step, we modify the probability distribution for each step of the random walk as follows:

$$\pi^d_{n_i n_{i+1}} = \begin{cases} \alpha & d(n_{i-1}, n_{i+1}) = 2 \\ 1 - \alpha & d(n_{i-1}, n_{i+1}) < 2 \end{cases} \tag{4}$$

where $\pi^d_{n_i n_{i+1}}$ represents the depth bias between nodes n_i and n_{i+1}, and $d(\cdot)$ denotes the shortest distance between two nodes. α is an adjustable hyper-parameter, when $\alpha > 0.5$, the biased random walk tends to traverse deeper paths.

Taking into account both semantic and depth biases, we ultimately employ the following formula to determine the probability of the random walk:

$$\pi_{n_i n_{i+1}} = \begin{cases} \alpha \cdot \beta_{n_i n_{i+1}} & d(n_{i-1}, n_{i+1}) = 2 \\ (1 - \alpha) \cdot \beta_{n_i n_{i+1}} & d(n_{i-1}, n_{i+1}) < 2 \end{cases} \tag{5}$$

Based on the aforementioned sampling probability, we can perform path extraction over the entire knowledge graph. Furthermore, we use the sequence $X = (e_1, r_2, e_3, r_4, \ldots, r_{t-1}, e_t)$ to represent the path extracted by our path extraction module, where e_i represents entities and r_i represents the relations between entities. Besides, we also use x_i to represent the ith element in the sequence.

Compared to the DeepWalk method defined by Eq. (1), which assigns the same sampling probability to each neighboring node, our path extraction module fully considers the semantic bias between nodes and focuses more on hard-to-predict difficult triples during the sampling process. Additionally, the guidance from depth bias allows the extracted paths to contain richer long-sequence information.

Guided by these two types of information, our path extraction module performs more efficient data preprocessing, providing higher-quality paths for the subsequent modeling process, thereby effectively improving the training efficiency of the model.

3.3. Text-augmented embedding module

Once the path extraction module identifies and extracts significant paths from the KG, the next step is to combine both the structural and textual information associated with these paths. This is crucial for enhancing the model's representation of entities and relations, especially when dealing with tasks that require both structural and textual understanding.

Text-augmented embedding module exploits GATs, PLMs, and feed-forward neural networks to effectively integrate structural and textual information. Specifically, we employ a GAT to aggregate the neighbor information for each entity, capturing the local structural information within the KG. For each entity e_i in the sequence X, we initialize $e^0 \in \mathbb{R}^d$ to represent the initial local structural embedding of the KG. Different neighbor entities should exert varying influence weights on the current entity. We employ the following function to calculate the influence weights of e_j on e_i:

$$f(e_i, e_j) = \frac{e_i W^Q \cdot (e_j W^K \odot r_{ij})^T}{\sqrt{d}} \tag{6}$$

where $W^Q, W^K \in \mathbb{R}^{d\times d}$, d is the vector dimensionality. The aforementioned formula can be regarded as the $QK^T/\sqrt{d}$ term in traditional attention mechanisms, and W^Q and W^K are trainable parameters. Feature vector similarity serves as a metric to measure the absolute influence among entities. The higher similarity between the feature vectors of a target entity and an adjacent entity suggests a stronger correlation, and the adjacent entity has a more significant influence in understanding the state and executing related tasks of the target entity. By calculating the dot product similarity, Eq. (6) effectively measures the absolute influence between entities.

Then we calculates the attention score $\alpha^l_{i,j}$ in the lth layer given the entity e_i and its 1-hop neighbors $e_j \in N_i$, where N_i denotes the set of all 1-hop neighbors of e_i:

$$\alpha^l_{i,j} = \frac{\exp(f(e^l_i, e^l_j))}{\sum_{e_{j'} \in N_i} \exp(f(e^l_i, e^l_{j'}))} \tag{7}$$

Where, e^l_i and e^l_j represent the embeddings of entities e_i and e_j at the l-th layer, respectively. Eq. (7) utilizes the *softmax* function to calculate local absolute influence from Eq. (6), transforming it into relative influence weights associated with the entity. Greater local absolute influence results in higher relative influence weights. The normalization process of *softmax* is particularly suited to the information aggregation in GATs.

Subsequently, we aggregate local neighbor features and update entity e_i's embedding using the following formula:

$$e^{l+1}_i = e^l_i + W_a \sum_{e_j \in N_i} \alpha^l_{i,j} e^l_i \odot r_{ij} \tag{8}$$

where W_a is designated as a trainable matrix that plays a critical role in dictating how features from neighboring entities are propagated to a given current entity. The GAT utilizes an attention function, crafted around local influence, to dynamically weigh and aggregate features from adjacent entities. This approach prioritizes features from more critical adjacent entities, minimizing noise interference and enhancing focus on important information, which improves the model's structural modeling capabilities.

To acquire additional textual semantic information, we utilize a prompt-based learning approach, employing the following

Table 1
Properties of FB15k-237 and WN18RR test sets.

Dataset	Shortest path distance			
	≥ 4	3	2	≤ 1
FB15k-237	0.8%	26.8%	72.4%	0.0%
WN18RR	34.3%	21.4%	9.3%	35.0%

prompt to process the textual descriptions of each entity and relation and encode them using PLMs, thereby acquiring textual information corresponding to each entity and relation.

$$\textit{The sentence}: \text{ ``}X\text{'' } \textit{means } [MASK] \tag{9}$$

where X denotes the text descriptions of entities or relations, and we take the hidden state h_{MASK} at the $[MASK]$ position as the text embedding vector X_T for entities and relations. Typical sentence embedding methods process all token embeddings within a sentence to generate a final representation. Variations in word frequency and form, such as capitalization, can influence these embeddings. However, prompt learning techniques infuse the full sentence's semantic information into one token's vector, ensuring that such variations do not directly affect the final text embedding. Additionally, by introducing prompt learning templates, the model's capacity to represent task-specific features is enhanced, leading to more purposeful and robust text embeddings.

Finally, We use two linear layers to process the two types of embedding vectors to obtain the final fusion embedding vector $\mathbf{x}_F$. The processes are summarized as follows:

$$\mathbf{x}_F = \text{Linear}(\mathbf{x}_S) + \text{Linear}(\mathbf{x}_T) \tag{10}$$

where $\mathbf{x}_S$ denotes the structural embedding vector captured using the GAT, and $\mathbf{x}_T$ represents the textual embedding vector learned through the PLM and prompt learning.

3.4. Attention-based recurrent network module

With the combined structural and textual embeddings in place, the model is now equipped with enriched representations for entities and relations. However, to effectively capture both short-term and long-term dependencies across multi-hop paths, it is essential to model sequential dependencies between these embeddings. This is where the Attention-based Recurrent Network (ARN) comes into play, allowing the model to attend to previous entities and relations adaptively, depending on the task at hand.

As shown in Table 1, FB15k-237 [32] has the most entities with the shortest 2-hop distances requiring greater long-term dependency model ability, while WN18RR [33] has a large proportion of entities with the shortest 1-hop distances. In addition, a 1-hop KGQA task requires more short-term knowledge, while multi-hop KGQA tasks require more long-term knowledge. Therefore, it is not sufficient to employ RNNs to capture long-term dependencies. Generic models must incorporate adaptive modules for different datasets and different tasks.

To capture both long-term and short-term dependencies adaptively, We propose an attention-based recurrent network, which utilizes an RNN to explicitly model multi-hop paths and employs cross-attention layers that allow the embedding of each relation to adaptively focus on the previous entities and relations. Besides, we denote the fusion vector $\mathbf{x}_F$ at recurrent step t as x_t, which serves as the input to the ARN network. We provide a detailed description of our ARN module in Fig. 3.

Specifically, given a sequence obtained by the path extraction module, we first use the text-augmented embedding module to obtain the corresponding embedding vector. We extract the sequence of relation from the whole sequence according to the type, i.e., we extract the sequence $X_r = (r_2, r_4, \ldots, r_{t-1})$ from the sequence $X = (e_1, r_2, e_3, \ldots, r_{t-1}, e_t)$. Then, we employ the scaled dot-product attention to compute the vector at each position. In a departure from conventional self-attention paradigms, we adopted the cross-attention strategy. We use $Q = W_q X_r$ as the query vector. By cross-attention, each entity's inference adaptively utilizes information from entities and relations in the previous recurrent steps, thus allowing our model to attend to different sequence length requirements. The inference for future entities can only leverage past information, so we use the masked strategy of the transformer to avoid the influence of future information.

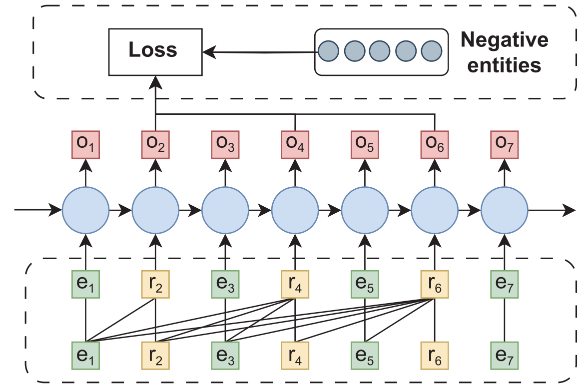

Fig. 3. We utilize cross-attention mechanisms to achieve adaptive attention to previous entities and relations. Subsequently, by employing a self-adversarial negative sampling loss, the model can learn knowledge of different lengths.

While self-attention mechanisms are effective, their efficiency is reduced by the sequential reasoning in LSTM models. Besides, it is common to predict the next tail entity based on past information and current relations, but it is not practical to predict the next relation this way. Relations help define the possible meanings for tail entities, but head entities or past information do not similarly limit the meanings of upcoming relations.

Later, we leverage Eq. (11) to obtain the hidden vector at each recurrent step:

$$h_t = \tanh(W_x x_t + W_h h_{t-1} + b) \tag{11}$$

where x_t denotes the input vector at the t step, h_{t-1} denotes the hidden vector at the $t-1$ step, b denotes the bias vector, W_x and W_h denotes the transfer matrix of the input and hidden vectors, respectively. The hidden vector at each position is calculated from the hidden vector at the previous position together with the input vector at the current position. It is worth noting that x_t in Eq. (11) can be either an entity's embedding vector or a relation.

3.5. Model learning

Although our model attempts to learn both local structural and textual semantic information in the text-augmented embedding module, the following two issues still negatively impact the learning of effective embeddings: (1) The aggregation of 1-hop neighbor entities' features may introduce noise, making it challenging for our model to learn robust representations. (2) A PLM, without specific fine-tuning, may not adequately capture the nuances of certain KG domains, and struggle to process texts with unconventional expressions. To overcome these challenges, we design two unique loss functions to enhance our model's robustness and adaptability, ensuring more accurate and domain-relevant representations in the KG.

Specifically, we design a masking and reconstruction process based on global influence weights, utilizing this reconstruction loss for self-supervised learning to obtain robust representations. Eq. (7) defines the influence weight of the triple (e_i, r_{rj}, e_j) for a single head entity e_i. By employing the *softmax* function, we can ascertain the relative importance of this triple compared to all other triples associated with this head entity. To effectively measure the global influence of this triple in the entire KG, we consider multiplying the number of triples corresponding to this head entity e_i, i.e., the degree $|N_i|$, with the softmax-normalized value from Eq. (7). This multiplication ensures that the influence of the triple is globally comparable, resulting in the following formula:

$$\omega(e_i, r_{ij}, e_j) = \frac{|N_i| \ \exp(f(e_i, e_j))}{\sum_{e_{j'} \in N_i} \exp(f(e_i, e_{j'}))} \tag{12}$$

By employing this method, we can select the top k triples with the highest importance scores for masking, that is, by deleting the edges of these triples to disconnect the head and tail entities:

$$\mathcal{M}_k = \{ \ (e_i, r_{ij}, e_j) \mid \omega(e_i, r_{ij}, e_j) \in \text{topk}(\mathcal{G}) \ \} \tag{13}$$

We denote the KG after masking as $\mathcal{G}_m$. Compared to random masking, masking and reconstruction based on influence scores enable our model to achieve higher learning efficiency and more effectively shield against the noise impacts caused by relatively less important triples.

For the reconstruction process, we first obtain the embedding vectors for each entity and relation using a GAT under the constraints of $\mathcal{G}_m$. We then utilize the following loss function to reconstruct the masked triples:

$$\mathcal{L}_{rec} = \sum_{(e_i, r_{ij}, e_j) \in \mathcal{M}_k} -\log \sigma(e_i^T \cdot (e_j \odot r_{ij})) \tag{14}$$

where $\sigma(\cdot)$ represents the sigmoid activation function, and this loss can be considered as the inverse process of the GAT's aggregation operation. The complete process of masking and reconstruction is illustrated in Fig. 4.

Masked triples are selected based on their global influence weights, allowing evaluation of how well the model's representations align with ground truth when key information is obscured. A higher $e_i^T \cdot (e_j \odot r_{ij})$ value indicates the model more accurately encodes significant triples, minimizing the reconstruction loss. This masking strategy aligns with the structural mask reconstruction strategy's goal to better model key information in knowledge graphs, minimize noise in dense areas, and emphasize important data in sparse regions, thereby enhancing the model's learning ability across varying structures and achieving more robust representations of knowledge graphs.

To address limitation (2) and enhance the generalization of the PLM across various datasets, we adopt a contrastive learning approach. Utilizing the triplet facts from knowledge graphs, we treat the text of the correct tail entity as the positive sample for the concatenated text of head entities and relations, and the text of incorrect tail entities as negative samples. Additionally, we use prompts with the same semantic meaning but different expressions to process the concatenated text of head entities and relations, treating it as the special positive sample for contrastive learning.

Specifically, We adopt the same prompt learning strategy as used in the text-augmented embedding module to obtain the embeddings e_{hr} for the head entity and relation, as well as the embedding e_t for the tail entity. Furthermore, we employ the prompt which, despite its varying form, retains the same semantic content to process the text resulting from the concatenation of head entities and relations. This process generates a particular positive sample e'_{hr} by PLM's encoding:

$$\textit{The sentence of ``X'' means } [MASK] \tag{15}$$

Then, we employ Eq. (2) to calculate the similarity between these two vectors. Subsequently, we use the following formula to compute the textual contrastive learning loss:

$$\mathcal{L}_{cl} = -\log \frac{\exp((sim(e_{hr}, e_t) + sim(e_{hr}, e'_{hr}))/\tau)}{\sum_{e_{t_i} \in T} \exp(sim(e_{hr}, e_{t_i})/\tau)} \tag{16}$$

where τ is a temperature parameter that controls the concentration level of the distribution. The T represents the negative samples, i.e., the incorrect triples (h, r, t_i). We adopt an in-batch strategy to generate negative samples, meaning that the size of T is batch size -1.

When PLMs effectively generalize to the domain knowledge of the KG, they can distinguish correct triples from erroneous triples, increasing the similarity score $sim(e_{hr}, e_t)$ for positive samples while diminishing it for negative samples $sim(e_{hr}, e_{t_i})$, thereby driving the $\mathcal{L}_{cl}$ loss towards zero. This optimization aligns with the objectives of textual contrastive learning. Additionally, PLMs maintain high similarity scores $sim(e_{hr}, e'_{hr})$ between pairs of positive samples that share semantics but differ in expression, which drives the $\mathcal{L}_{cl}$ loss towards zero. This optimization allows the model to focus on semantic content over formal differences, enabling it to adeptly handle diverse texts with unconventional expressions like *Which software does [E]'s employer provide technical support for?* in the KGQA task.

We employ the self-adversarial negative sampling technique to assign different weights to each negative sample and utilize a negative sample-based loss function as our core loss for model optimization, efficiently learning long-term relation dependencies within the KG. Specially, we use the following distribution for self-adversarial negative sampling:

$$p(y, y'_j) = \frac{\exp \alpha(y \cdot y'_j)}{\sum_i \exp \alpha(y \cdot y'_i)} \tag{17}$$

where y denotes the true input embedding x_t at each step, y' denotes the negative sample embedding and α is the temperature of the negative sampling process. High-similarity negative samples, or hard negatives, pose significant challenges for model discrimination. The self-adversarial negative sampling strategy we employ assigns weights to negative samples based on their similarity to the current node. This weighting mechanism prioritizes hard negatives during the training process, thereby enhancing the model's recognition capabilities for such samples.

According to ARN, the output vector h_{t-1} at the previous recurrent step $t-1$ is the prediction for the input x_t. Our goal is to make h_{t-1} as similar as possible to the positive sample x_t at the next position and as dissimilar as possible to the negative samples. Therefore, We apply the following function on the sequence $X = (e_1, r_2, e_3, \ldots, e_T)$ as our core loss function:

$$\begin{aligned}\mathcal{L}_m = -\sum_{t=1}^{T} \Big(& \log \sigma(h_{t-1} \cdot y_t) \\ & + \sum_{j=1}^{n} p(y_t, y'_j) \log \sigma(-h_{t-1} \cdot y'_j) \Big)\end{aligned} \tag{18}$$

where h_{t-1} denotes the hidden vector obtained through the ARN at previous recurrent steps $t-1$, while y_t and y' respectively denote the input embedding of the true and negative samples at current recurrent steps t.

Ultimately, we employ the following joint loss function to optimize our model:

$$\mathcal{L} = \mathcal{L}_m + \lambda_1 \mathcal{L}_{rec} + \lambda_2 \mathcal{L}_{cl} \tag{19}$$

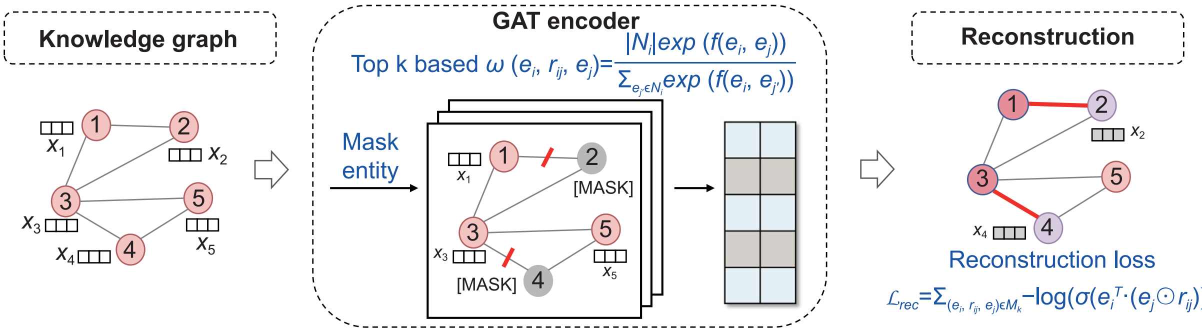

Fig. 4. We employ the global influence weight $\omega(e_i, r_{ij}, e_j)$ to effectuate a mask operation on important entities within the KG. Following the embedding process via a GAT, we reconstruct the masked entities by reconstruction loss $\mathcal{L}_{rec}$, thereby establishing a more robust representation of the local structure.

Table 2
Statistics of the KGC datasets. The numbers of entities and relations are indicated, respectively, by #E and #R. #Train, #Valid, #Test denote the number of facts triples in the training set, validation set, and test set, respectively.

Dataset	#E	#R	#Train	#Valid	#Test
FB15k-237	14,541	237	272,115	17,535	20,466
WN18RR	40,943	11	86,835	3,034	3,134
YAGO3–10	123,182	37	1079,040	5,000	5,000
Wikidata5M-Trans	4, 594, 485	822	20, 614, 279	5,163	5,163
Wikidata5M-Ind	4, 579, 609	822	20, 496, 514	6,699	6,894

Table 3
Statistics of the question answering datasets. #Train, #Valid, #Test indicate the number of question and answer pairs.

Dataset	#Train	#Valid	#Test
MetaQA 1-hop	96,106	9,992	9,947
MetaQA 2-hop	118,948	14,872	14,872
MetaQA 3-hop	114,196	14,274	14,274
WebQSP	2,998	100	1,639

where λ_1 and λ_2 represent the weights for the structural reconstruction loss and the textual contrastive learning loss, respectively, and we set them as parameters that can be updated during model training.

4. Experiments

In this section, we will discuss the details of the experiments, including the experimental setup, overall experiments result in KGC and KGQA tasks, and analysis of each module of TAARN and its capabilities.

4.1. Experimental setup

4.1.1. Datasets

The KGC dataset is made up of triples, some of which also include text corpus related to entities and relations. The QA dataset contains an associated KG and a QA dataset containing some question-and-answer statements. Each question in the QA dataset is made of natural language containing entities in the KG, and the answers are entities in the KG.

For the KGC task, we use four datasets: Wikidata5M [34], YAGO3-10 [35], FB15k-237 [32] and WN18RR [33]. Since YAGO3-10 lacks textual corpus, we only utilize the textual information contained in the KG. Notably, Wikidata5M is configured in two settings: the transductive and inductive settings. In the transductive setting, there exists a possibility of overlap between entities in the training and test sets. Conversely, the inductive setting is designed such that the entities in the test set are entirely exclusive from those in the training set, presenting a significantly more rigorous challenge, particularly in terms of the model's ability to infer and generalize effectively on unseen entities. Table 2 shows the statistics for these datasets.

For the QA task, we use a large-scale dataset, MetaQA [28], and a smaller-scale dataset, WebQuestionSP [36]. Compared to MetaQA, WebQuestionSP contains fewer question-answer pairs but requires a larger KG for answering. This setup thoroughly tests the model's ability to capture structural information as well as its capability to integrate structural and textual information. Additionally, to verify the ability of the model to perform multi-hop reasoning and implicit knowledge learning, we conduct experiments by randomly discarding 50% of the triples. Similar to YAGO3-10, MetaQA, and WebQuestionSP do not have a complex text corpus apart from the text carried by the labels. However, some models utilize the text from WikiMovies [37] and Wikipedia [38] as an additional textual supplement. The statistics of the datasets are shown in Table 3.

4.1.2. Baselines

We perform a comprehensive comparison with the current state-of-the-art models. Specifically, for KGC tasks, we report two levels of graph-based models on three datasets. The triple-level models include RotatE [5], TransE [6], ComplEx [14],ConvE [33], HAKE [18], CAKE [39], HousE [40] while the graph-level models include PTransE [41], RSN [10], CompGCN [42], Rethinking [43], SE-GNN [44]. Additionally, we compare our models with recent text-based models such as SimKGC [7], KG-BERT [16], MTL-KGC [45], and StAR [46]. For the QA task, we compare TAARN with the five models reported in EmbedKGQA [26], which include VRN [47], KV-Mem [37], GraftNet [24], PullNet [23] and EmbedKGQA [26].

4.1.3. Implementations

We put our model TAARN into practice with the PyTorch 1.13 framework [48]. Besides, we employ the widely-used *bert-base-uncased* model from Hugging Face, as it has been observed that better PLMs can yield improved results. In addition, Adam optimizer is employed to optimize our model and we dynamically adjust the learning rate with the OneCycleLR schedule. For the hyperparameters, we set the α of the path extraction module to 0.7. We adjust the temperature of negative sampling within {0.5, 1.0}. The tuning range for the learning rate is {0.0001, 0.0003, 0.0005}. All experiments are conducted on Tesla V100 GPUs.

In QA tasks, questions are presented in natural language. Due to the flexibility of natural language, it is impractical to generate a relation label for every text as a relation. Therefore, for both KG and QA training, we use the textual embedding vector $\mathbf{x}_T$ of relations or questions as the input for ARN. This approach allows for a unified model training, integrating KG knowledge into QA tasks and standardizing the process across both tasks.

Table 4
KGC results obtained on the FB15k-237, WN18RR and YAGO3-10 datasets.

Model	FB15k-237				WN18RR				YAGO3-10			
	MRR	H@1	H@3	H@10	MRR	H@1	H@3	H@10	MRR	H@1	H@3	H@10
Graph-based methods												
TransE	0.330	0.231	0.369	0.528	0.223	0.014	0.401	0.529	0.160	0.103	–	0.279
PTransE	0.290	0.203	–	0.451	0.340	0.272	–	0.464	0.190	0.122	–	0.323
ComplEx	0.323	0.229	0.353	0.513	0.468	0.427	0.485	0.554	0.360	0.260	0.400	0.550
ConvE	0.325	0.237	0.356	0.501	0.430	0.400	0.440	0.520	0.440	0.350	0.490	0.620
RotatE	0.338	0.241	0.375	0.533	0.476	0.428	0.492	0.571	0.495	0.402	0.550	0.670
HAKE	0.346	0.250	0.381	0.542	0.497	0.452	0.516	0.582	0.545	0.462	0.596	0.694
RSN	0.270	0.192	–	0.418	0.400	0.380	–	0.448	0.240	0.164	–	0.373
CompGCN	0.355	0.264	0.390	0.535	0.479	0.443	0.494	0.546	–	–	–	–
SE-GNN	0.365	0.271	0.399	0.549	0.484	0.446	0.509	0.572	–	–	–	–
Rethinking	0.355	0.264	0.389	0.535	0.472	0.437	0.485	0.544	–	–	–	–
CAKE	0.321	0.226	0.355	0.515	–	–	–	–	–	–	–	–
HousE	0.361	0.266	0.399	0.551	0.511	0.465	0.528	0.602	–	–	–	–
Text-based methods												
KG-BERT	–	–	–	0.420	0.216	0.041	0.302	0.524	–	–	–	–
MTL-KGC	0.267	0.172	0.298	0.458	0.331	0.203	0.383	0.597	–	–	–	–
StAR	0.296	0.205	0.322	0.482	0.401	0.243	0.491	0.709	–	–	–	–
SimKGC	0.336	0.249	0.362	0.511	0.666	0.587	0.717	0.800	–	–	–	–
Ours	**0.389**	**0.294**	**0.408**	**0.572**	**0.689**	**0.623**	**0.724**	**0.824**	**0.588**	**0.512**	**0.628**	**0.721**

"–" means to remove the module.
Bold represents the best results.

4.2. Overall result

4.2.1. Knowledge graph completion results

Experiments demonstrate that TAARN outperforms existing graph-based and text-based KGC models, as shown in Table 4. The key reasons behind TAARN's superior performance lie in its ability to effectively integrate both local structural information and textual information, while most baseline models focus on either structural or textual aspects independently. TAARN's novel combination of a GAT based on local influence and a PLM-based prompt learning mechanism allows it to capture fine-grained local structural nuances and semantic information from text simultaneously. This balanced fusion of structural and textual data gives TAARN a distinct advantage, especially in datasets like WN18RR and YAGO3-10, where both types of information are critical.

The high density of FB15k-237 [7] places demands on the model to capture complex structural information. TAARN stands out in this regard, utilizing a GAT method based on local influence, which surpasses baseline models in handling complex local structures and effectively capturing critical local structural information. Unlike simpler GNN-based models like CompGCN and SE-GNN, which primarily aggregate neighborhood information without considering the relative influence of entities, our GAT based on local influence prioritizes more influential neighboring entities and relations. This selective aggregation helps the model filter out noisy neighbor information, leading to more accurate predictions, particularly on FB15K-237, where multi-hop reasoning is crucial.

Furthermore, TAARN employs a mask-reconstruction strategy based on global influences that effectively reduces noise during training, focusing learning on the most predictive structural features, thereby enhancing the model's robustness in capturing complex structural information. This is particularly beneficial for datasets like FB15K-237, which contain noisy and redundant relations. By masking less relevant parts of the KG during training, TAARN learns to focus on the most informative connections, improving both its generalization and reasoning capabilities, and leading to better performance than models like TransE and HAKE, which lack such noise reduction mechanisms.

These modules endow TAARN with superior structural capturing capabilities, enabling it to achieve the best results on the FB15K-237 dataset.

However, this approach also introduces additional computational overhead during both the training and inference phases, which can become a bottleneck, particularly in real-time applications or when dealing with large-scale KGs with millions of entities. This trade-off between robustness and computational efficiency is something that should be taken into account when considering TAARN for deployment in time-critical or resource-constrained environments.

Furthermore, on the WN18RR dataset, TAARN reveals its capability to integrate structural and textual information. Wang et al. [7] point out that prior textual information serves as an excellent complement for WN18RR, enabling models that capture textual information to achieve good performance on this dataset. The TAARN model leverages prompt learning to activate the powerful text representation capabilities of PLM, effectively eliminating potential embedding biases. In addition, the integration of textual contrastive learning loss, particularly through prompt-based positive sample generation techniques, further enhances the model's ability to handle new domain knowledge and various textual expressions, which in turn facilitates an effective acquisition and processing of textual semantic information. On top of that, TAARN can capture the information within triples and exhibit higher efficiency in propagating path information, thus performing better on WN18RR. The WN18RR dataset primarily involves 1-hop and 4-hop reasoning tasks, which require the model to both efficiently capture direct relationships and propagate information across longer paths. TAARN's ARN module, which adaptively focuses on the most relevant multi-hop paths, gives it an edge over models like RotatE and ComplEx, which struggle with longer path dependencies. This ability to dynamically adjust its focus based on the task at hand allows TAARN to excel in both short- and long-hop reasoning.

Besides, Table 1 analyzes the shortest distances between the entities and shows different datasets have different properties. Specifically, the FB15K-237 test set features more 2-hop and 3-hop reasoning tasks, whereas the WN18RR dataset emphasizes 1-hop and 4-hop reasoning capabilities. The SOTA results on both FB15K-237 and WN18RR datasets with different characteristics, demonstrate our model's adaptive attention capability towards different sequence lengths.

4.2.2. Question answering over knowledge graph results

KGQA datasets utilize a graph-structured KG for natural language-based question answering, thus providing an effective

Table 5
KGQA results obtained on the original dataset and under 50% KG settings.

Model	MetaQA KG-Full			MetaQA KG-50			WebQSP KG-Full	WebQSP KG-50
	1-hop	2-hop	3-hop	1-hop	2-hop	3-hop		
EmbedKGQA	97.5	98.8	**94.8**	83.9	91.8	70.3	66.6	53.2
VRN	97.5	89.9	62.5	–	–	–	–	–
KV-Mem	96.2	82.7	48.9	63.6(75.7)	41.8(48.4)	37.6(35.2)	46.7	32.7(31.6)
GraftNet	97.0	94.8	77.7	64.0(91.5)	52.6(69.5)	59.2(66.4)	66.4	48.2(49.7)
PullNet	97.0	**99.9**	91.4	65.1(92.4)	52.1(90.4)	59.7(85.2)	68.1	50.1(51.9)
Ours	**97.9**	**99.9**	94.1	**86.4**	**92.3**	**74.9**	**72.4**	**56.9**

"–" means to remove the module.
Bold represents the best results.
The data representation in () refers to the results obtained by these models after incorporating the additional corpus from the WikiMovies and Wikipedia datasets.

measure for model proficiency in capturing and integrating structural and textual information. As shown in Table 5, owing to TAARN's excellent capabilities in modeling and integrating structural and textual information, it exhibits exceptional performance on both the MetaQA and WebQSP datasets. Specifically, the GAT based on local influence effectively captures local structural information, while the PLM-based prompt learning exhibits strong text representation capabilities. This synergy is particularly evident in the KGQA task, where questions are presented in natural language and require both structural and textual comprehension. Models like EmbedKGQA and PullNet either focus heavily on structural retrieval or text matching, but TAARN's combination of GAT and PLM allows it to fluidly switch between these types of reasoning, leading to superior performance on both MetaQA and WebQSP datasets. The model's strong performance on the 3-hop tasks in MetaQA, for example, highlights its capability to handle complex multi-hop reasoning while maintaining high accuracy in simpler 1-hop tasks.

Furthermore, the contrastive learning strategy employs prompt-based positive sample generation techniques to bring unconventional question statements with the same semantic information closer to conventional text descriptions in the feature space. This effectively captures the semantic associations between the question semantics and the main and answer entities, enhancing the TAARN model's ability to handle knowledge graph question-answering tasks. Then, the synergy of mask-reconstruction strategy and contrastive learning integrate robust and generalizable structural and textual representations, resulting in SOTA performance.

Various inference hops are indicative of the model's generalizability. We utilize the ARN module to enhance multi-hop performance without jeopardizing the efficiency of the 1-hop performance. The effectiveness of this module outshines that of the EmbedKGQA, which primarily relies on utilizing local structural information. In contrast, TAARN's ARN module adaptively focuses on the most relevant paths, improving its ability to handle both short and long reasoning paths. This is particularly advantageous in datasets like WebQSP, where the sparsity of the KG necessitates more efficient path reasoning. As a result, TAARN achieves substantial improvements in the 2-hop and 3-hop tasks, while maintaining competitive performance in 1-hop tasks.

Experiments under the 50% setting, which results in a more incomplete KG and longer inference path, demonstrate our model's proficiency in multi-hop reasoning on incomplete KGs. Models that perform subgraph retrievals such as KV-Mem, GraftNet, and PullNet show significant performance degradation when dealing with incomplete KGs. In contrast, the model based on KG representation learning can learn implicit knowledge from the existing KGs to implicitly complete the missing information and achieve better performance. However, EmbedKGQA only uses ComplEx to learn triple-level knowledge from KG, in contrast, our model has better structural modeling capability to capture longer path information adaptively.

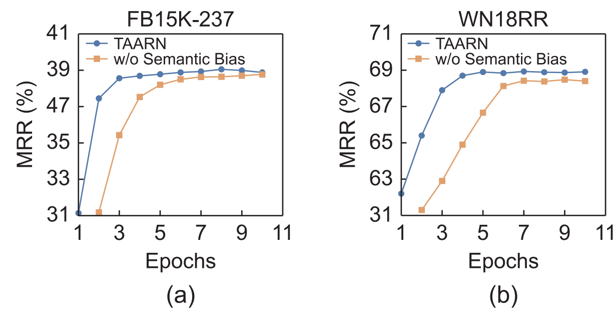

Fig. 5. MRR(%) results on the (a) FB15K-237 and (b) WN18RR w.r.t epochs.

Moreover, the WebQuestionSP dataset which has fewer question-answer pairs and a larger KG, favors models with a more efficient capability to capture textual information and effectively identify crucial structural information. We enhance the generalizability of the model through the use of prompt learning and contrastive learning. These techniques allow TAARN to better align textual information with structural data, which is essential for datasets like WebQSP that have fewer question-answer pairs but a larger KG. By using contrastive learning to pull related question statements closer in the feature space, TAARN effectively learns to generalize across different textual expressions of the same underlying question, leading to improved performance over models like GraftNet and PullNet, which lack such a mechanism for handling semantic variation in text. Moreover, by employing GAT based on local influence and adopting a mask-reconstruct strategy that focuses on globally important entities within the KG, our model can disregard more noise, resulting in more robust performance.

4.3. Analysis

To validate the effectiveness of each module within our model, we conduct a series of experiments to analyze our model comprehensively.

4.3.1. Sensitivity to random walk

We conduct two types of experiments on the KGC task to observe the impact of semantic bias in random walks and the extracted path lengths on the model's performance.

Fig. 5 demonstrates the impact of semantic bias on model training. It is evident that the model's convergence speed significantly slows down when semantic bias is removed on two datasets. This phenomenon can be attributed to the adoption of semantic bias for path extraction, effectively and efficiently utilizing the prior knowledge in the PLM to identify complex and challenging samples, thereby enhancing training efficiency.

Fig. 6 explores the influence of varying the length of extracted paths on the model's performance. Notably, on the FB15K-237

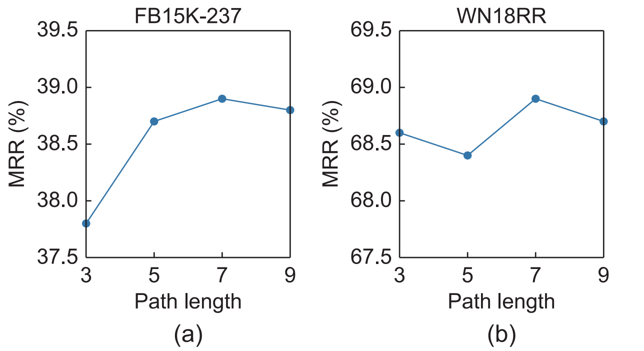

Fig. 6. MRR(%) results on the (a) FB15K-237 and (b) WN18RR w.r.t path length.

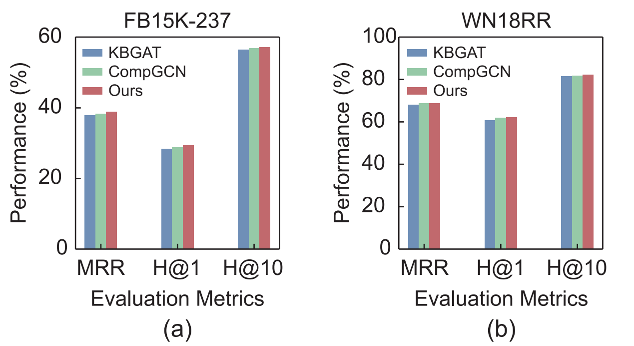

Fig. 7. Effectiveness of graph attention network. (a) FB15K-237. (b) WN18RR.

dataset, our model exhibits improved performance as the path length increases. As shown in Table 1, FB15K-237 displays a higher proportion of 2-hop and 3-hop instances. The explicit modeling of multi-hop paths enables our model to effectively learn information from these paths, contributing to enhanced overall performance. For the WN18RR dataset, which contains a significant number of 1-hop and greater than 4-hop instances, our model maintains consistently stable performance. This suggests that the ARN module in our model adeptly recognizes crucial information in diverse datasets (such as 1-hop in WN18RR and 2-hop/3-hop in FB15K-237). Despite the introduction of additional noise with longer path lengths, the model demonstrates robustness by avoiding a decline in performance.

4.3.2. Effectiveness of graph attention network

To investigate the impact of the GAT based on local influence on model performance, this section replaces the GAT part of the proposed TAARN model with the graph neural networks from KBGAT and CompGCN, conducting experiments on two KGC datasets. The results, shown in Fig. 7, indicate a decrease in model performance when the GATs from KBGAT and CompGCN replace the GAT based on local influence. KBGAT utilizes feature concatenation and linear transformation $c_{h,r,t}^{(k)} = W_1^{(k)}\left[X_h^{(k)} \parallel X_t^{(k)} \parallel X_r^{(k)}\right]$ to obtain joint feature vectors and compute attention scores:

$$\alpha_{h,r,t}^{(k)} = \frac{\exp(LeakyReLU(W_2^{(k)} c_{h,r,t}^{(k)}))}{\sum_{(r',t')\in N_h} \exp(LeakyReLU(W_2^{(k)} c_{h,r',t'}^{(k)}))} \tag{20}$$

However, this feature aggregation $X_h^{(k+1)} = g\left(\sum_{(r,t)\in N_h} \alpha_{h,r,t}^{(k)} c_{h,r,t}^{(k)}\right)$ overlooks the structured knowledge organized in triples within the KG, resulting in inefficient information propagation. In contrast, the GAT based on local influence, designed with the multiple relational characteristics of KGs in mind, efficiently aggregates information from adjacent entities by calculating local influence

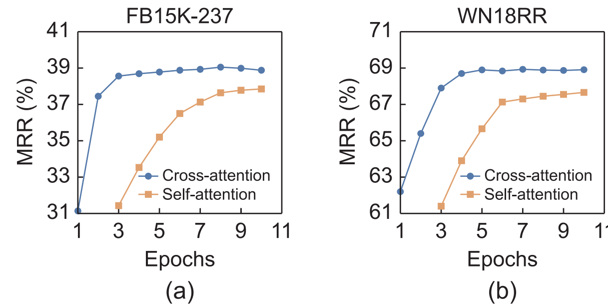

Fig. 8. Effectiveness of cross-attention. (a) FB15K-237. (b) WN18RR.

between entities and filtering noise, thus enhancing focus on key structural information and improving model performance. Moreover, CompGCN effectively aggregates neighboring node information:

$$X_t^{(k+1)} = g\left(\sum_{(h,r)\in N_t} W_{\lambda(r)}^{(k)} \phi\left(X_h^{(k)}, X_r^{(k)}\right)\right) \tag{21}$$

This method considers interactions between entities and relations, which is more efficient than KBGAT but still neglects the influence and feature interactions between entities, hence a decline in model performance when replaced by the CompGCN model. These experimental results validate the effectiveness of the GAT based on local influence.

4.3.3. Effectiveness of cross-attention

This section examines the influence of cross-attention layers on the performance of the TAARN model by substituting its cross-attention mechanism with a self-attention algorithm across two datasets for KGC. The findings, illustrated in Fig. 8, reveal that models equipped with self-attention layers exhibit a marked slowdown in convergence and a substantial decline in overall performance. The TAARN model, which sequentially predicts the next entity, utilizes cross-attention to integrate prior relational data into current predictions, providing a more substantial role for earlier entities and relations than achieved with self-attention. This integration significantly increases the utilization efficiency of prior information, thereby boosting the model's training efficacy and performance. These results underscore the superior functionality and validity of the cross-attention mechanism in enhancing model performance in complex tasks.

4.3.4. Unseen entity inference analysis

To verify the generalizability of our model, we conduct KGC experiments on the Wikidata5M dataset, which offers both transductive and inductive experimental setups. reveal that our model outperforms the baselines in both settings.

The large-scale KG and abundant text descriptions of Wikidata5M challenge the model's ability to model complex structures and integrate structural and textual information. Our model, utilizing masked reconstruction loss, effectively focuses on important knowledge within large and complex structures, demonstrating strong structural modeling capabilities. Employing prompt learning and contrastive learning, our model efficiently represents text descriptions. By integrating these two types of information, our model achieves SOTA results. The inductive setting is our primary focus, where entities in the test set do not appear in the training set, posing a challenge to the model's generalization capabilities. Leveraging a PLM to incorporate additional textual information mitigates the challenge posed by unseen entities, as the inherent prior knowledge within the PLM enhances the

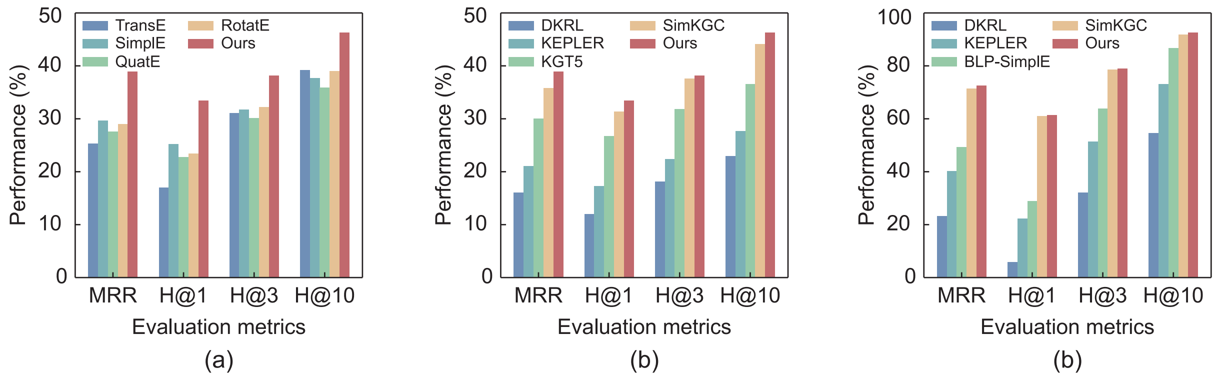

Fig. 9. KGC results on the Wikidata5M datasets. (a) Transductive setting 1. (b) Transductive setting 2. (c) Inductive setting.

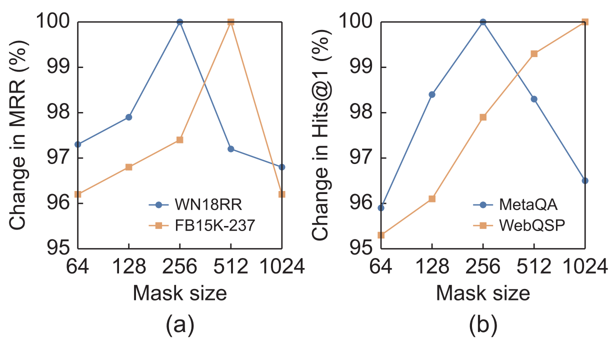

Fig. 10. Change in MRR(%) w.r.t masking size. (a) WN18RR and FB15K-237. (b) MetaQA and WebQSP.

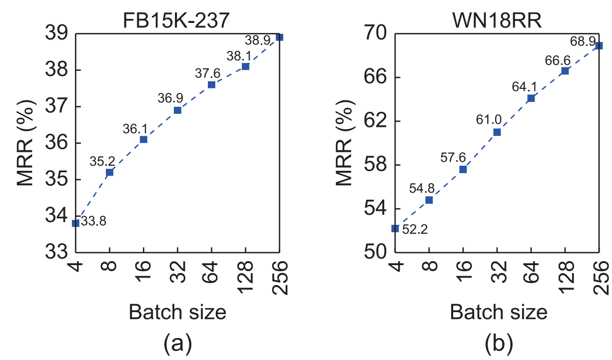

Fig. 11. MRR(%) results on the (a) FB15K-237 and (b) WN18RR w.r.t batch size.

generalizability of text-based models, despite limited training on structural information from the KG. Our model, employing prompt learning and contrastive learning, demonstrates more efficient textual representation capabilities, outperforming baseline models in addressing unseen entity challenges. Furthermore, information propagation and aggregation of neighboring entity features through a GAT allow our model to exhibit strong inductive performance. The well-trained attention matrices in the model can effectively generalize to these unseen entities. And the exceptional performance in this setting showcases its robust generalization ability.

4.3.5. Sensitivity to masking size

Our exploration into the model's sensitivity to the masking size k hyperparameter reveals intriguing findings, as depicted in Fig. 10. We discern clear preferences for optimal masking size settings, contingent on the unique attributes of each dataset.

In the KGC task, a nuanced distinction emerges between datasets: for WN18RR, a masking size of 256 is identified as the most effective, while FB15K-237 demonstrates better performance with a masking size of 512. This variation in optimal settings suggests a correlation with dataset characteristics, particularly concerning data sparsity. Similarly, in the KGQA task, the MetaQA dataset responds best to a masking size of 256, contrasting with WebQuestionSP, which favors a larger masking size of 1024.

We postulate that this variance is linked to the degree of sparsity in the datasets. In the case of sparser datasets, such as FB15K-237 and WebQuestionSP, a larger masking size appears beneficial in augmenting the model's ability to capture and represent the data effectively. However, for denser datasets, such as WN18RR and MetaQA, the model already possesses a strong learning capacity. Excessively large masking sizes in these contexts could lead to an overshadowing of existing data richness, potentially causing information loss and adversely affecting the model's performance. This insight into the relation between dataset sparsity and optimal masking size underscores the importance of tailored hyperparameter tuning in achieving the best model performance across varying knowledge graphs.

4.3.6. Sensitivity to batch size

Fig. 11 offers an in-depth examination of how the MRR metric responds to variations in batch size. The experiments reveal a noteworthy trend: the model's performance exhibits an upward trajectory as the batch size is incrementally increased. This basis for the enhancement can be attributed to our model's employment of an in-batch negative sample generation technique in both the textual contrastive learning and core loss components. As the batch size expands, each positive sample within the batch is contrasted against a progressively larger pool of negative samples. Consequently, it improves its ability to distinguish correct from incorrect information, which leads to a more robust and effective learning outcome.

4.3.7. Ablation study

We conduct a comprehensive experimental evaluation, wherein each module within the model is independently removed, to meticulously observe the influence of each module on the overall performance of the model. Table 6 presents the results of the ablation study conducted on the KGC and KGQA tasks.

For the KGC task, the removal of the semantic bias module, which utilizes prior knowledge in PLMs to prioritize challenging samples in path extraction, results in only a minor decrease in performance. This aligns with its role in enhancing training efficiency and marginally improving model performance. The depth bias introduces more long-sequence semantic information. On the

Table 6
Results of the ablation study.

Model	KGC						KGQA					
	FB15K-237			WN18RR			MetaQA KG-Full			MetaQA KG-50		
	MRR	H@1	H@10	MRR	H@1	H@10	1-hop	2-hop	3-hop	1-hop	2-hop	3-hop
TAARN	**0.389**	**0.294**	0.572	**0.689**	**0.623**	**0.824**	**97.9**	**99.9**	**94.1**	**86.4**	**92.3**	**74.9**
-Semantic bias	0.387	0.287	**0.573**	0.684	0.617	0.819	97.6	**99.9**	93.9	86.2	91.4	73.2
-Depth bias	0.384	0.286	0.562	0.686	0.621	0.823	97.8	99.2	92.3	85.8	91.2	71.6
-Embedding Prompt	0.386	0.289	0.566	0.635	0.574	0.742	97.1	99.3	93.2	85.7	92.0	72.8
-Positive Sampling	0.387	0.289	0.570	0.651	0.571	0.737	97.8	99.2	92.3	85.8	91.2	71.6
-Contrastive Loss	0.388	0.292	0.568	0.649	0.568	0.722	97.4	99.6	92.9	85.3	92.1	73.5
-Text-Augmented	0.384	0.289	0.572	0.573	0.512	0.683	–	–	–	–	–	–
-GAT	0.373	0.279	0.558	0.678	0.596	0.812	96.2	98.9	93.5	83.8	90.2	71.1
-Rec Loss	0.380	0.286	0.568	0.664	0.587	0.808	96.0	99.5	92.1	84.5	91.3	72.4
-Cross-Attention	0.376	0.287	0.560	0.673	0.599	0.816	97.5	98.3	90.3	83.9	90.8	70.7

"–" means to remove the module.

WN18RR dataset, which relies more on 1-hop and greater than 4-hop semantic knowledge, the removal of this module does not result in a loss of 1-hop information, hence the slight performance decline. Conversely, the FB15K-237 dataset depends more on 2-hop and 3-hop reasoning capabilities; removing the deep bias guidance results in a significant loss of this semantic knowledge and a relatively larger performance decline.

The omission of the embedding prompt, prompt-based positive samples and contrastive loss modules does not significantly affect the model's performance on FB15k-237, attributable to its reliance on structural modeling capacity over text representation enhancements. However, for WN18RR, which demands more textual information, the absence of the embedding prompt, prompt-based positive samples, and contrastive loss modules leads to performance degradation due to reduced sentence embedding ability and generalization of textual information by the PLM. Furthermore, when all text-augmented modules, including all parts involving the use of PLM and textual information, are removed, the structural representation remains unaffected, maintaining stable performance on FB15K-237 but underperforming on WN18RR due to the absence of additional textual information.

The utilization of the GAT based on local influence is pivotal for capturing local structural information efficiently. Its elimination results in a marked performance reduction on the FB15K-237 dataset, validating our analysis. Similarly, the reconstruction loss module enhances the robustness of representation learning by utilizing structural information, particularly on the WN18RR dataset which chiefly uses 1-hop information, where it mitigates multi-hop noise impact. The absence of this module leads to performance degradation due to increased noise interference. The attention module plays a crucial role in adaptively focusing on varied information lengths, and its removal leads to diminished performance across both datasets.

For KGQA tasks, where questions are posed in text, the model's text capturing ability is critically evaluated. Further, answering these text-form questions within a graph structure tests the model's proficiency in synthesizing structural and textual information. The TAARN model addresses this task by removing the main entity from the question, transforming the question into an unconventional text format, and treating it as a relation within the KG. The representation of such unconventional text formats significantly impacts the model's performance on KGQA. The embedding prompt, prompt-based positive samples, and contrastive loss modules enhance the TAARN's capability to represent various textual expressions. Removing these three modules impairs the model's ability to express unconventional text, leading to significant performance drops on two datasets. Moreover, the TAARN model's approach to handling KGQA relies on the PLM's ability to capture textual information, making it challenging to perform ablation studies that remove text-augmented modules.

Experiments categorized by inference length validate the impact of different model components on adaptive multi-hop reasoning performance. Removing the depth bias guidance reduces the model's ability to acquire long-sequence semantic knowledge, causing more pronounced performance declines in 2-hop and 3-hop tasks than in 1-hop tasks. Furthermore, after removing the GAT based on local influence, the model's ability to process local structural information is compromised, particularly affecting performance more in 1-hop reasoning tasks than in multi-hop tasks. The complexity of multi-hop reasoning tasks underscores that removing attention modules affects 2-hop and 3-hop reasoning tasks more significantly, emphasizing the module's crucial role in enhancing the model's multi-hop processing capabilities.

5. Conclusion

In this paper, we propose TAARN, a novel knowledge representation model that combines graph attention networks, pre-trained language models, and recurrent neural networks with cross-attention mechanisms. TAARN adapts to tasks such as knowledge graph completion (KGC) and knowledge-intensive question answering (QA) by integrating structural and textual information, and leveraging a semantic-depth guided path extraction algorithm to capture long-term dependencies. Our experiments demonstrate that TAARN outperforms existing models on KGC benchmarks and delivers competitive results on QA tasks. However, the model's complexity introduces higher computational costs, which may limit its applicability in resource-constrained or real-time environments. Future work will focus on developing more efficient variants of TAARN, optimizing its scalability, and reducing computational overhead, particularly for large-scale and real-time applications.

CRediT authorship contribution statement

Quntao Zhu: Methodology, Conceptualization. **Mengfan Li:** Software, Methodology. **Yuanjun Gao:** Resources. **Yao Wan:** Project administration. **Xuanhua Shi:** Supervision, Methodology. **Hai Jin:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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